

Clear-Claim — Product Overview

A hardware-gated, clear-signed reward claimer for DePIN operators.

In one line

Clear-Claim lets an AI agent manage a DePIN operator's reward claims autonomously, while the operator keeps final, **readable** approval on their Ledger device — so the agent does the work, but the human authorizes exactly what they can see.

The problem

DePIN operators (Helium, Hivemapper, DIMO, and similar) run always-on nodes that earn rewards continuously. To claim those rewards today, they choose between two bad options:

- **Auto-claim with a hot key** stored in a config file — copyable by malware, exposed to phishing, and a standing target. This is the pattern behind multiple real 2025–2026 drains of agent and operator wallets.
- **Blind-sign a raw hash** — the operator taps "approve" on an unreadable string like `0x23b872dd0000...`, with no way to verify the amount or destination. A large share of the \$3B stolen in 2025 exploited exactly this gap between *signing* and *understanding*.

Neither gives the operator autonomy **and** safety. That's the gap Clear-Claim fills.

How it works



1. **The agent watches** the operator's claimable balance and the current gas price.
 2. **The agent decides** — it only proposes a claim when the reward clears a floor and gas is below the operator's ceiling. This is the "intelligence" layer.
 3. **The agent assembles** the claim transaction — but holds no private key.
 4. **The Ledger displays it in plain language** — thanks to a clear-signing descriptor, the operator reads `(Claim 142.5 RWRD to 0x9aF3...)` on the device, not a hash.
 5. **The operator approves on-device.** The key never leaves the hardware.
 6. **The CLI broadcasts** the signed transaction.
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What makes it different

Most "AI agent + crypto" tools stop at *gating a transaction behind hardware*. But a gate is meaningless if the device shows an unreadable hash — the human is still blind-signing a blank check. Clear-Claim's core idea is that **the human-in-the-loop is only real if the transaction is readable**. It pairs an autonomous agent with clear signing so the approval step actually protects the operator.

The same descriptor that makes the claim readable is also a contribution back to Ledger's open clear-signing registry — turning the project from a tool that *uses* Ledger into one that *extends* its ecosystem.

Who it's for

- **DePIN operators** running one or many nodes who want hands-off reward management without parking a hot key in a config file.
 - **DePIN-as-a-service hosts** who operate nodes for others and need a trust boundary between running the node and moving the money.
 - **Any builder** putting an autonomous agent near real value who wants the agent to act freely but never to move funds the human hasn't read and approved.
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Built with

- **Solidity** — the DePIN reward distributor (EVM testnet).
- **Node + TypeScript** — the agent decision loop, transaction assembly (viem), and broadcast.
- **Ledger DMK (Device Management Kit)** — drives the on-device signing in-process, so the agent genuinely uses the DMK rather than shelling out to a CLI (demoable on Speculos, no

hardware required).

- **ERC-7730 clear signing** — the descriptor that makes the claim human-readable on-device.
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The thesis, in one sentence

The missing layer in agentic crypto isn't an agent that can sign — it's an agent whose signing the human can actually verify.